

Application Architecture

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md §2 · HLD: high-level-design.md

The solution consists of **four business applications sharing a common backend** (freeze §2).

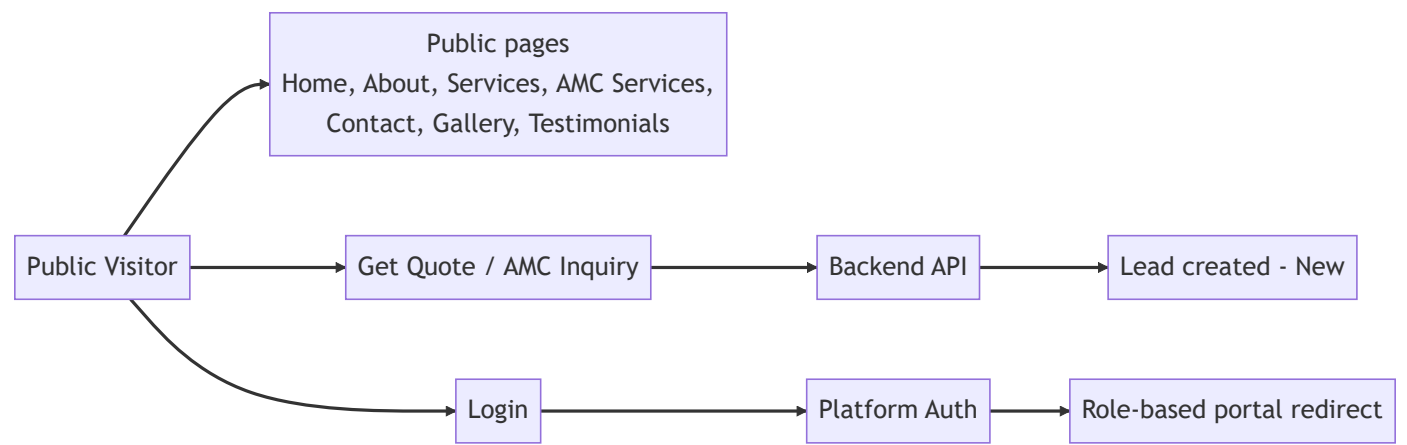
1. Public Website

Aspect	Detail
Domain	www.aarvii.in
Audience	Public Visitor (anonymous)
Purpose	Business showcase, lead generation, AMC plan information, contact inquiries, quote requests
Content	Reuse existing public website content wherever possible (freeze §2)

Pages (approved): Home · About Us · Services · AMC Services · Contact Us · Gallery · Testimonials · Login **Enhancements (approved):** Get Quote · AMC Inquiry

Behavior:

- Get Quote / AMC Inquiry / Contact submissions call the backend and **automatically create leads** (freeze §10), firing the Lead Created notification (§17).
- Login routes authenticated users to their portal (Customer / Engineer / Admin) based on role.
- No authenticated functionality lives on the public site itself.



2. Customer Portal

Aspect	Detail
Channels	Web (React SPA module) + Customer App (Flutter)
Audience	Customer

Aspect	Detail
Features (freeze \$2)	Dashboard · AMC Details · Service History · Upcoming Visits · Tickets · Invoices · Profile Management · Password Reset

Behavioral rules:

- Shows only the customer's **own** data (freeze \$3).
- AMC view shows the **current active term**; renewal history is admin-only (§8).
- Service history shows **approved** visit reports only (§13).
- Tickets: create, track, and **reopen closed tickets** (§14).
- Invoices: view and **download PDF** (§16).
- AMC renewal can be **requested** from the portal (§3).

3. Engineer Portal

Aspect	Detail
Channels	Web (React SPA module) + Engineer App (Flutter, offline-capable)
Audience	Engineer
Features (freeze \$2)	Assigned Visits · Assigned Tickets · Visit Reporting · Photo Upload · GPS Capture · Selfie Capture · Customer Signature · Ticket Creation

Behavioral rules:

- Work queue limited to **assigned** visits and tickets (§3, §15).
- Visit completion enforces the mandatory evidence checklist: selfie, GPS (lat/long/timestamp), ≥1 photo, customer signature, remarks (§12).
- Photos categorized Before / During / After; video upload supported (§12, §15).
- Submitted reports go to **admin review** before customer visibility (§13).
- Engineers can create tickets during visits (§3) but **cannot** manage customers, plans, or contracts (§15).

4. Admin Portal

Aspect	Detail
Channel	Web (React SPA module)
Audience	Admin
Features (freeze \$2)	Lead Management · Customer Management · Site Management · Asset Management · AMC Plans · AMC Contracts · Scheduling · Ticket Management · Engineer Management · Invoice Management · Reporting

Behavioral rules:

- Full lead pipeline management and conversion (Customer + Site + Initial AMC Contract, §10).
- AMC contract master + term management with full renewal history visibility (§8); versioned plan management (§9).
- Visit scheduling: auto-generation oversight, rescheduling, **mandatory engineer assignment** (§11).
- Visit report **review & approval** gate (§13).
- Invoice lifecycle management with PDF generation (§16, §19).

5. Backend APIs

Aspect	Detail
Host	Ashraak.Api — .NET 10 modular monolith composition root (shared by all four applications)
Core (frozen)	Auth, Tenant, Users, Files, Notifications (email), Audit, Webhooks, ApiKeys, Caching/Outbox
CCTV business modules (new)	Lead, Customer/Site/Asset, AMC (Plans + Contracts), Scheduling/Visits, Tickets, Engineer, Invoices, Reporting — registered at Host Layer 2
API style	REST (Minimal APIs), versioned routes, JWT bearer auth; OpenAPI feeds the mobile SDK
Cross-module rule	Business modules → Core only via <code>SharedKernel.Contracts</code> ; no Core edits (freeze \$20)

Platform contracts consumed:

Contract	Used for
Auth / RBAC / ABAC	Login, roles (Customer/Engineer/Admin), permission checks
IFileStorage	Photos, videos, selfies, signatures, generated PDFs
INotificationService	Email notifications; SMS via provider integration
IAuditService (observer)	Automatic business-event audit capture
IWebhookPublisher	Outbound CCTV catalog events
ICacheService / Outbox	Caching and reliable event publication

6. Database

Logical view only — **physical design and ERD are deferred to D0-4** per D0 quality rules.

Store	Usage
PostgreSQL	CCTV business data in new per-module schemas (platform convention); core platform schemas remain untouched
Redis	Platform caching, sessions, rate limiting
MongoDB	Platform audit entries
File storage (local/S3/Azure)	Binary content via platform Files module

Logical data areas implied by the freeze (to be modeled in D0-4): Leads · Customers · Sites · Contact Persons (≤3/site) · Asset Summaries · AMC Plans (versioned) · AMC Contracts + Terms · Scheduled Visits · Visit Reports/Evidence · Tickets · Engineers · Invoices · Notifications log.

7. External integrations

Integration	Direction	Purpose	Freeze ref
SMTP / email provider	Outbound	Email notifications	\$17

Integration	Direction	Purpose	Freeze ref
SMS gateway	Outbound	SMS notifications, Login OTP, Password Reset OTP	\$17
Cloud storage (S3/Azure, optional)	Outbound	File blobs via platform Files	\$12, \$19, \$20
Webhook consumers	Outbound	Platform webhook delivery to external systems	\$20
App stores	Release	Customer & Engineer app distribution	\$18

Explicitly excluded (freeze \$21): WhatsApp integration, payment gateway, accounting/GST systems.

Related documents

- [high-level-design.md](#)
- [module-architecture.md](#)
- [mobile-architecture.md](#)
- [navigation-map.md](#)